

WASP-36b

R-1644

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220101 · created 2026-08-02T07:12:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459580.8308 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.049
Transit depth (Rp/Rs)^2	2.71 +/- 1.61 [%]
Orbital Inclination (inc)	86.03 +/- 2.85
Airmass coefficient 1 (a1)	0.128 +/- 0.063
Airmass coefficient 2 (a2)	1.26 +/- 0.53
Scatter in the residuals of the lightcurve fit is	39.45 %
Best Comparison Star	None
Optimal Aperture	2.77
Optimal Annulus	3.0
Transit Duration (day)	0.086 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.049 vs 0.1368 (deviation 0.58σ)
Duration fit vs published	0.086 d vs 0.0758 d (deviation 1.02σ)
Mid-time O–C	8.7 min
Depth SNR	3.4
Residual scatter	39.45 %

Light curve

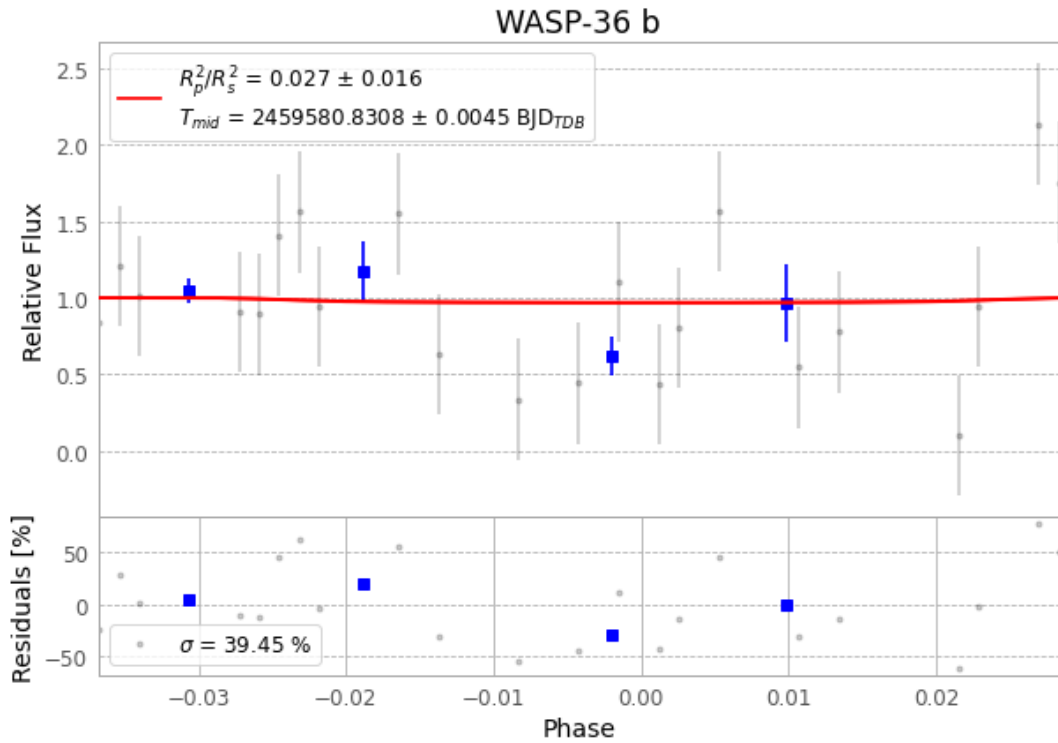


Figure 1 — FinalLightCurve_WASP-36 b_2021-12-31.png (SHA-256 5d7bcfac993ce5ed...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [617, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 722722f893a1f604...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-36220101055713.FITS → WASP-36220101085113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220101.log (6db9853c1c11...), FinalLightCurve_WASP-36 b_2021-12-31.png (5d7bcfac993c...), FinalLightCurve_WASP-36 b_2021-12-31.csv (a1225b02c918...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:12Z.